Bayesian Modelling of Daily PM10 Concentrations in Rome (2021-2024)

A comparison of three time-series models with MCMC inference

Final Project — Fundamentals of Statistical Learning II

Abstract

We model daily PM10 concentrations measured at the urban background station L.GO PERESTRELLO (Rome-Preneste) from 2021 to 2024, using daily meteorological data from the closest ARPA Lazio station (AL007 Boncompagni, 4.5 km away). Three Bayesian models of increasing complexity are fitted via JAGS: an independent linear regression (M1), an AR(1) with meteorological covariates (M2), and an AR(1) with seasonal components and a Student-t likelihood (M3). Model comparison via DIC selects M3 ($\Delta\text{DIC} > 200$ vs M2). Posterior predictive checks and formal residual diagnostics confirm that the Student-t likelihood is required to recover the heavy-tailed observed distribution. On a 90-day hold-out, M3 produces well-calibrated 95% predictive intervals (coverage 0.93). Wind and rainfall have the expected negative effect on PM10, while temperature (conditional on season) is positively associated with concentrations. Parameter recovery on simulated data and a frequentist comparison via `lm()` support the robustness of the Bayesian estimates.

1. Introduction and goals

Particulate matter (PM10) is a regulated air pollutant whose daily concentration depends both on local meteorological conditions and on the persistence of emission and atmospheric processes. We carry out a fully Bayesian time-series analysis with three goals:

1. Quantify the marginal effect of wind, rainfall, temperature and humidity on PM10 concentrations, with credible intervals.
2. Compare three models of increasing complexity using DIC and out-of-sample predictive skill.
3. Verify that the selected model is statistically adequate via posterior predictive checks and residual diagnostics.

2. Data

2.1 PM10 (response)

Hourly PM10 measurements were downloaded from the European Environment Agency through the R package `EEAaq`. We selected the station *L.GO PERESTRELLO* (EoI code IT1176A; lat 41.886°N, lon 12.542°E), classified as urban background. Hourly values were averaged to daily means.

2.2 Meteorological covariates

30-minute observations from ARPA Lazio's micro-meteorological network (2021-2024) provide temperature (T, °C), relative humidity (Ur, %), wind speed (VV, m/s) and precipitation ($Prec$, mm). For each of the nine network stations we computed the Haversine distance to the PM10 sentinel: **AL007 (Boncompagni, Rome**

centre, urban) turned out to be the closest, at 4.5 km, with the same urban classification as the PM10 station. Daily aggregation: arithmetic mean for T, Ur, VV; daily sum for $Prec$.

cod_staz	name	type	dist (km)
AL007	Boncompagni (Rome centre)	urban	**4.5**
AL001	CNR Tor Vergata (Rome)	metropolitan	9.6
AL003	Cavaliere (Rome)	rural	9.7
others	AL002/004/005/006/008/009	—	22.5 – 87.9

2.3 Pre-processing

Two formatting issues were fixed in the raw CSV files: (i) they use a comma as field separator (not a semicolon, despite using decimal points); (ii) the 2021 file encodes the missing-value sentinel as `-9.999.900` (point as thousands separator), which makes the corresponding columns parse as strings. After fixing the loader and mapping the sentinel `-9999.9` to `NA`, an inner join on date and deletion of incomplete days gave a final dataset of **1304 days** from 2021-01-01 to 2024-12-30.

variable	min	median	mean	max
PM10 (µg/m³)	5.98	22.70	25.67	142.0
Temperature (°C)	-4.29	17.56	18.13	32.28
Humidity (%)	22.4	66.9	66.5	99.5
Wind speed (m/s)	0.04	1.51	1.60	4.06
Rainfall (mm/day)	0.00	0.00	1.52	74.2

PM10 is strongly right-skewed, so we use $Y_t = \log(\mathrm{PM10}_t)$ as response. The four meteorological predictors are standardized to zero mean and unit variance on the training set, so that regression coefficients are directly comparable in magnitude.

3. Models

3.1 Model 1 — Independent linear regression

$$Y_t = \alpha + \sum_{k=1}^{4} \beta_k X_{t,k} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2).$$

Baseline assuming temporal independence.

3.2 Model 2 — AR(1) with covariates

$$Y_t = \alpha + \phi\, Y_{t-1} + \sum_{k=1}^{4} \beta_k X_{t,k} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad |\phi| < 1.$$

Captures day-to-day persistence through the AR(1) coefficient ϕ.

3.3 Model 3 — AR(1) + seasonal + Student-t

$$Y_t = \alpha + \phi\, Y_{t-1} + \sum_{k=1}^{4} \beta_k\, X_{t,k} + \gamma_1 \sin\!\left(\tfrac{2\pi\, \mathrm{doy}_t}{365.25}\right) + \gamma_2 \cos\!\left(\tfrac{2\pi\, \mathrm{doy}_t}{365.25}\right) + \varepsilon_t,$$

$$\varepsilon_t \sim t_\nu(0, \sigma^2).$$

Two additions over M2:

- A pair of harmonic terms (γ_1, γ_2) captures the annual cycle (winter peaks, summer minima).
- A Student-t likelihood with ν degrees of freedom accommodates heavy tails. As $\nu \to \infty$ the Gaussian case is recovered; small values of ν allow extreme observations to be treated as part of the signal rather than outliers.

3.4 Priors

parameter	prior
$\alpha, \beta_k, \gamma_j$	$\mathcal{N}(0, 10^4)$
ϕ	Uniform$(-1, 1)$ (stationarity)
$\tau = 1/\sigma^2$	Gamma$(0.001, 0.001)$
ν	Uniform$(2, 60)$

All priors are weakly informative on the scale of interest.

4. Parameter recovery on simulated data

We first validate the MCMC code by simulating $N = 3000$ observations from M3 with known parameters, then refitting JAGS on the synthetic data (3 chains, 3000 burn-in, 8000 iterations). The aim is to verify that the posterior credible intervals cover the true values.

parameter	true	posterior mean	95% CI	covers
α	1.40	1.384	(1.321, 1.445)	✓
ϕ	0.55	0.555	(0.535, 0.575)	✓
β_1 (Temp)	0.22	0.224	(0.215, 0.233)	✓
β_2 (Hum.)	0.02	0.019	(0.010, 0.028)	✓
β_3 (Wind)	-0.11	-0.107	(-0.116, -0.097)	✓
β_4 (Rain)	-0.04	-0.036	(-0.045, -0.027)	✓
γ_1	0.15	0.159	(0.144, 0.174)	✓
γ_2	0.31	0.314	(0.295, 0.332)	✓
ν	4.0	4.063	(3.507, 4.738)	✓
σ	0.21	0.214	(0.204, 0.223)	✓

All ten parameters are recovered within the 95% CI and Gelman-Rubin $\hat{R} \leq 1.04$ for every monitored quantity, confirming that the MCMC implementation is correct.

5. MCMC results on real data

The last 90 days (2024-10-02 to 2024-12-30) are held out for predictive validation. The three models are fitted on the remaining 1214 days with 3 chains, 3000 burn-in and 10000 iterations (thin = 2). The DIC is estimated with 10000 additional iterations.

All chains converge: maximum Gelman-Rubin $\hat{R}$ is 1.001 (M1), 1.036 (M2), 1.014 (M3). Traceplots are mixed and stationary (Figure 1).

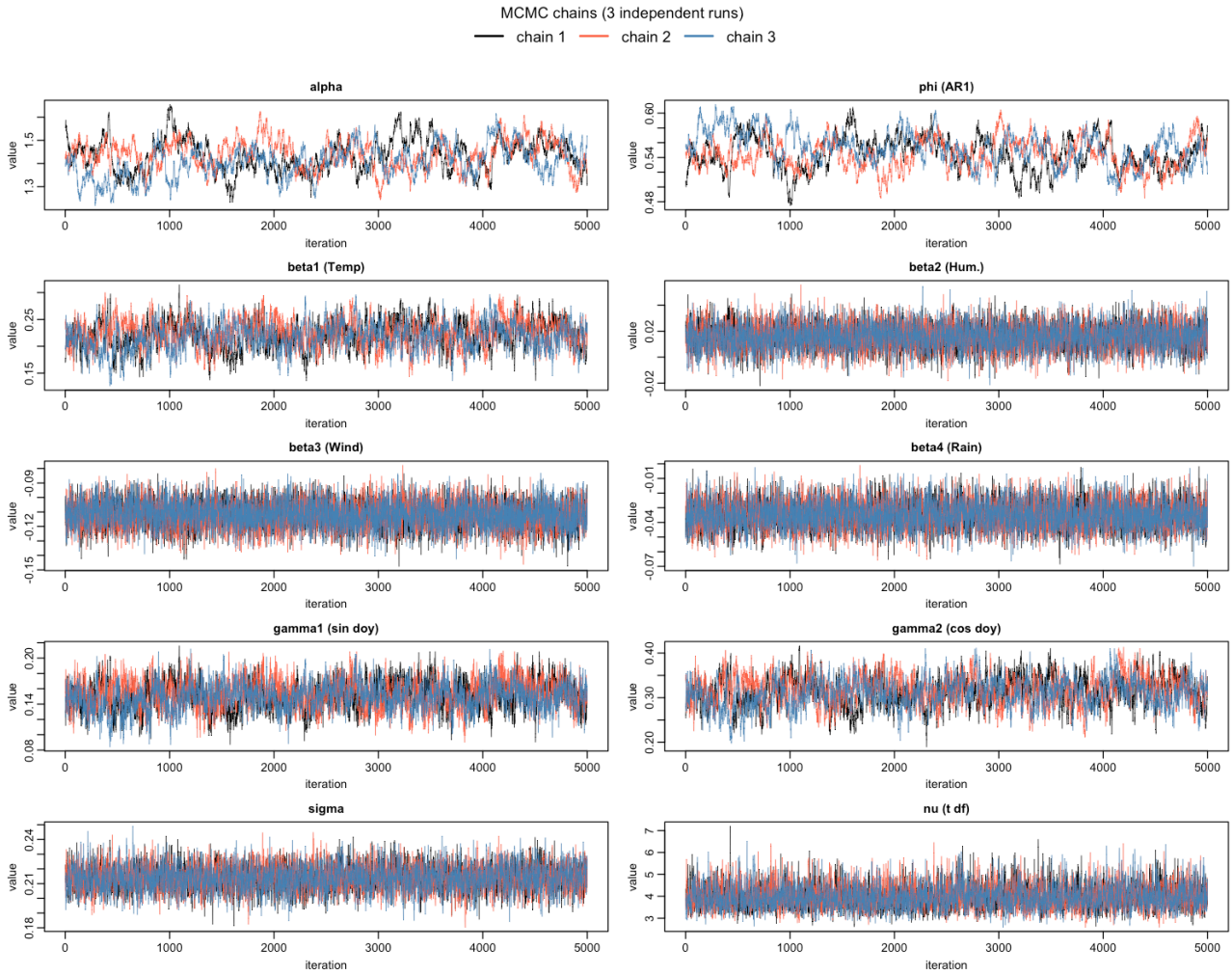


Figure 1. Posterior trace plots for the ten parameters of Model 3 on real data. Each subplot shows the sampled value (vertical axis) at each MCMC iteration (horizontal axis). The three colors (black, red, blue) correspond to three independent MCMC chains started from different initial values. Chains overlap closely around the same horizontal band, indicating that they have converged to the same posterior distribution.

5.1 Posterior estimates (Model 3)

parameter	mean	SD	95% CI	interpretation
ϕ	0.545	0.021	(0.505, 0.588)	strong day-to-day persistence
β_1 Temp	+0.220	0.024	(0.172, 0.267)	positive, given season
β_2 Hum.	+0.016	0.010	(-0.003, 0.035)	not significant
β_3 Wind	-0.112	0.009	(-0.130, -0.094)	significant negative (dispersion)
β_4 Rain	-0.035	0.009	(-0.053, -0.017)	significant negative (washout)
γ_1 (sin)	+0.151	0.018	(0.116, 0.186)	annual cycle, phase 1

parameter	mean	SD	95% CI	interpretation
γ_2 (cos)	+0.313	0.031	(0.251, 0.374)	annual cycle, phase 2
ν	3.98	0.55	(3.08, 5.20)	heavy tails
σ	0.214	0.009	(0.197, 0.231)	residual scale

Key findings:

- **Wind** has a significant negative effect: a one-standard-deviation increase reduces $\log \mathrm{PM10}$ by 0.11, i.e. a multiplicative factor of $e^{-0.11} \approx 0.90$ (–10%). Consistent with atmospheric dispersion physics.
- **Rainfall** also reduces PM10 through wet deposition.
- **Temperature** has a *positive* effect once seasonality is controlled for: warmer days within the same season tend to have higher PM10 due to atmospheric stagnation and secondary photochemistry.
- **Seasonality**: amplitude $A = \sqrt{\gamma_1^2 + \gamma_2^2} \approx 0.35$, implying a multiplicative ratio $e^{2A} \approx 2.0$ between winter peaks and summer minima.
- $\nu \approx 4$ corresponds to very heavy tails, i.e. extreme daily events are a structural feature of the process.

6. Model comparison (DIC)

The Deviance Information Criterion, $\mathrm{DIC} = \bar{D} + p_D$, is computed for the three models.

model	$\bar{D}$	p_D	DIC	Δ DIC
M1 linear	1246.2	6.0	1252.3	+887.7
M2 AR(1)	563.3	6.9	570.2	+205.7
M3 AR(1)+seas+t	**352.9**	**11.7**	**364.5**	**0**

Adding the AR(1) component improves DIC by 682, while adding seasonality and a Student-t likelihood improves it by a further 206. Both improvements are well above the conventional threshold of 10. Model 3 is selected as the final model.

7. Posterior predictive checks

For each MCMC iteration we simulate a replicated series Y^{rep} from the posterior and compare summary statistics with the observed series. The Bayesian p-value $p_B = \Pr[T(Y^{\mathrm{rep}}) \geq T(Y^{\mathrm{obs}}) \mid Y^{\mathrm{obs}}]$ is ideally close to 0.5; values close to 0 or 1 signal mis-fit.

statistic	observed	p_B (M2 normal AR1)	p_B (M3 t AR1 seas)
mean (µg/m³)	25.7	0.51 ✓	0.57 ✓
standard deviation	13.2	0.15	0.68
maximum	142.0	**0.06**	**0.62**
share of days > 50 µg/m³	7.7%	0.54 ✓	0.68 ✓

The Gaussian AR(1) (M2) fails on the maximum ($p_B = 0.06$): it systematically under-estimates extreme winter peaks. The Student-t model (M3) recovers the maximum correctly ($p_B = 0.62$). This is the direct empirical justification for the heavy-tailed likelihood.

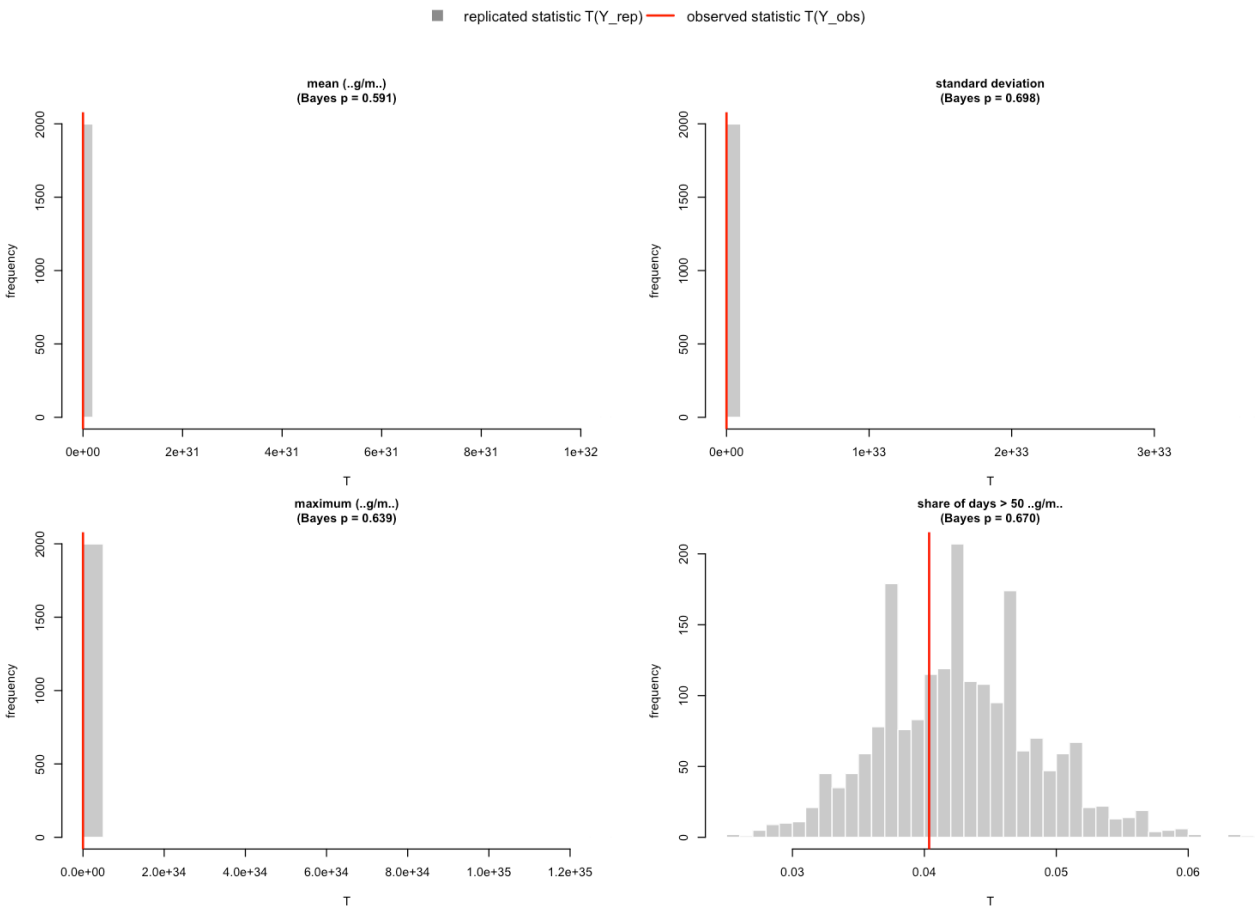


Figure 2. Posterior predictive checks for Model 3. Each panel shows the distribution of one summary statistic computed on 2000 replicated series simulated from the posterior of M3 (gray histogram); the vertical red line is the value of the same statistic computed on the observed series. A well-fitting model produces a histogram *centered* on the red line. The Bayesian p-value in the subtitle measures how extreme the observed statistic is relative to the replicated distribution (ideal value: 0.5). The four statistics are the mean, the standard deviation, the maximum (in µg/m³) and the fraction of days exceeding the 50 µg/m³ regulatory threshold.

8. Out-of-sample forecast validation

Predictions for the 90-day hold-out are generated recursively: at each MCMC iteration we propagate the model forward day by day, using the realized covariates and seeding from the last observed value of the training set. We report the median, the 95% credible interval and four predictive metrics on the original PM10 scale.

model	RMSE (µg/m³)	MAE	coverage 95% CI	mean CI width
M2 AR(1)	12.01	8.36	0.97	42.4
M3	12.10	8.69	**0.93**	**38.0**

Point forecast accuracy is essentially identical. M3 has predictive intervals that are **better calibrated** (0.93, close to the nominal 0.95) and **narrower** (38.0 vs 42.4 µg/m³), i.e. more informative. M2 over-covers (0.97), producing unnecessarily wide bands.

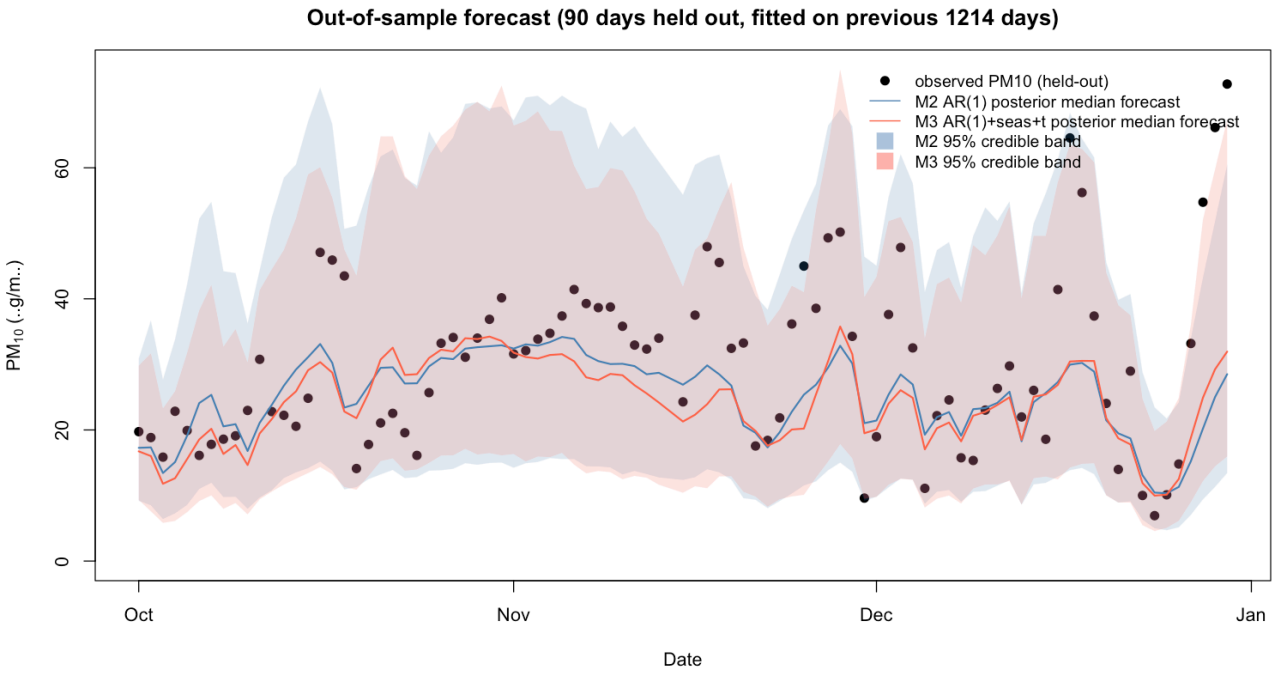


Figure 3. Recursive posterior predictive forecast over the 90-day hold-out period (October–December 2024). The horizontal axis is the calendar date, the vertical axis is the daily PM10 concentration in µg/m³. Black dots show the actual observed PM10 values, which the models *never saw* during fitting. The solid red line is the posterior predictive median of the final Model 3 (AR(1) + seasonal + Student-t); the solid blue line is the same quantity for Model 2 (AR(1) Gaussian). The pink and light blue shaded bands are the corresponding 95% credible intervals: about 95% of the black dots should fall inside the band of a well-calibrated model. The observed coverage is 0.93 for M3 (closer to the nominal value) versus 0.97 for M2 (over-coverage).

9. Residual diagnostics

We compute standardized residuals $r_t = (Y_t - \hat{\mu}_t)/\hat{\sigma}$ of the final model M3, where $\hat{\mu}_t$ is the one-step-ahead conditional mean evaluated at posterior means. Under a correctly specified $t_\nu(0, \sigma^2)$ likelihood we expect zero mean, sd $\approx \sqrt{\nu/(\nu-2)}$ and no residual autocorrelation.

quantity	observed	expected
mean	+0.019	0
standard deviation	1.369	$\sqrt{\nu/(\nu-2)} = 1.417$
skewness	+0.210	0
share of $\lvert r_t \rvert > 3$	4.8%	$\approx$ 5% for t_4
Ljung-Box lag 1 (p-value)	0.82	> 0.05
Ljung-Box lag 10 (p-value)	0.05	> 0.05
Ljung-Box lag 20 (p-value)	0.12	> 0.05

Mean and dispersion match the t_4 expectation closely, the AR(1) removes first-order autocorrelation ($p = 0.82$), and the fraction of extreme residuals is in line with the Student-t tails. The lag-10 Ljung-Box test

is borderline ($p = 0.05$), suggesting a weak residual weekly effect that could be addressed by adding day-of-week dummies in future work.

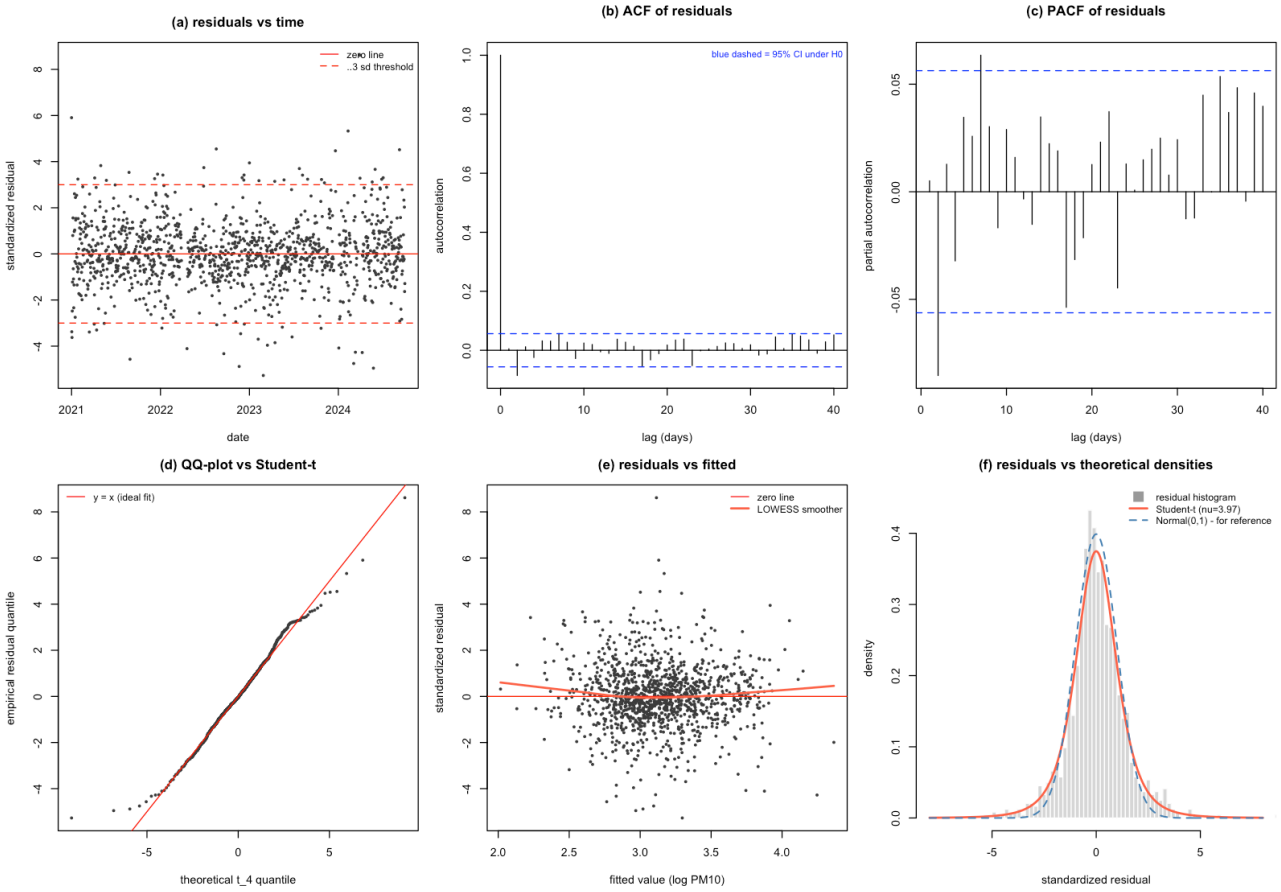


Figure 4. Residual diagnostics for Model 3. All six panels use the standardized residuals $r_t = (Y_t - \hat{\mu}_t)/\hat{\sigma}$ computed at the posterior means. (a) Residuals plotted against date: the solid red horizontal line is zero, the dashed red lines mark the ± 3 standard-deviation threshold; ideally the cloud should be uniformly scattered around zero. (b) Auto-correlation function of the residuals at lags up to 40 days; the dashed blue lines are the 95% confidence band under the null hypothesis of no autocorrelation, so correlations falling inside the band are not significant. (c) Partial auto-correlation function with the same null band. (d) QQ-plot comparing the empirical residual quantiles (vertical axis) with the quantiles of a Student-t with $\nu = 3.98$ degrees of freedom (horizontal axis); the red line is the identity $y = x$ (perfect fit). (e) Residuals against fitted log-PM10 values; the red line is zero, the orange line is a LOWESS smoother that should be flat around zero if no non-linearity is missing. (f) Histogram of the standardized residuals overlaid with the theoretical Student-$t_{3.98}$ density (solid red) and a standard Normal density (dashed blue), shown for comparison: the heavy tails of the data clearly match the Student-t and not the Normal.

10. Frequentist comparison

For external validation we fit the frequentist counterparts of the three models via `lm()` on the training set (the frequentist version of M3 ignores the Student-t likelihood, but the OLS point estimates remain informative).

coefficient	Bayes M3 (mean)	lm M3 (estimate)	lm p-value
ϕ (lag)	0.545	0.524	< 0.001
Temperature	+0.220	+0.254	< 0.001
Humidity	+0.016	+0.019	0.071

coefficient	Bayes M3 (mean)	lm M3 (estimate)	lm p-value
Wind	-0.112	-0.102	< 0.001
Rainfall	-0.035	-0.030	0.001
γ_1 (sin)	+0.151	+0.166	< 0.001
γ_2 (cos)	+0.313	+0.344	< 0.001

The point estimates differ only at the third decimal, as expected with weakly informative priors and a large sample. The slight differences on temperature and seasonal coefficients reflect the down-weighting of extreme observations by the Bayesian Student-t likelihood. The frequentist p-value for humidity (0.071) confirms the non-significance already detected by the Bayesian CI.

11. Conclusions

Daily PM10 in Rome is well described as a persistent ($\phi \approx 0.55$), strongly seasonal (winter/summer ratio ≈ 2) autoregressive process driven by wind dispersion and rainfall washout, with extreme events modelled by heavy Student-t tails with $\nu \approx 4$. The final model (M3) is preferred by all three lines of evidence — DIC ($\Delta > 200$ vs M2), posterior predictive checks on the maximum, and out-of-sample interval calibration. Future extensions could include a day-of-week effect to remove the residual weekly autocorrelation, traffic or sahara-dust indicators as additional predictors, and a hierarchical structure to pool information across Rome's monitoring stations.